

1 A Computational Reproducibility Check of Hurst and Kavanagh (2017): Life History
2 Strategies and Psychopathology

3 Jiyueyi Wang¹

4 ¹ School of Psychology, Nanjing Normal University

5 Author Note

6 This manuscript reports a computational reproducibility check of Hurst and
7 Kavanagh (2017) conducted with the authors' openly archived data. The replication was
8 written in R and the report was prepared with papaja. All replication code is available
9 from the author upon request.

Abstract

Computational reproducibility—obtaining the same reported numbers from the same data with independently written code—is a minimal requirement for credible quantitative research. Using the openly archived supplementary data of Hurst and Kavanagh (2017, *Evolution and Human Behavior*; $N = 138$), who reported that faster life history strategies accompany more psychopathology, greater aggression, and poorer attachment, I independently re-computed all six tables of the original article in R and compared every reported value, cell by cell, against the published value. Across 196 paired comparisons, 89.3% were reproduced exactly and a further 9.2% differed only by last-digit rounding or a single significance star; all 64 statistical inferences were consistent with the original. Two Mini-K partial correlations could not be reproduced from the data, but because the matching HKSS coefficients reproduced exactly and two independent partial-correlation methods agreed, these are most plausibly transcription errors in the original article rather than replication failures. Three discrepancies that initially made two tables look irreproducible were traced to my own misreading of how the raw data were coded—scoring origin, yes/no coding, and column order—and disappeared after consulting the codebook. The original results are computationally robust; the only real limitation to full transparency was the absence of shared analysis code.

Keywords: computational reproducibility, life history theory, psychopathology, open data, replication

Word count: Approx. 3,600

A Computational Reproducibility Check of Hurst and Kavanagh (2017): Life History Strategies and Psychopathology

Introduction

How do early-life experiences become adult mental-health outcomes? Hurst and Kavanagh (2017) approached this question through life history theory, which treats childhood adversity not merely as trauma but as an environmental cue that shapes a person's developmental strategy along a slow-fast continuum. In a community sample ($N = 138$), they reported that faster life history strategies—indexed by the Mini-K and the High-K Strategy Scale (HKSS)—were associated with more symptoms of psychopathology, greater aggression, and poorer perceived parental attachment, whereas links with self-harm frequency, family structure, and participant sex were weak or non-significant.

This paper does not re-evaluate those substantive claims. It asks a narrower, foundational question: given the authors' openly archived data, can the values printed in the article be regenerated with independently written analysis code? This property—*computational reproducibility*—is distinct from replicability with new data and is increasingly treated as a minimum standard for transparent quantitative research (Hardwicke et al., 2018; Nosek et al., 2022). Because the original design is cross-sectional and correlational, with all results expressed as descriptive statistics, correlations, and regressions, every reported number should in principle be exactly recoverable from the public data. I therefore reproduced all six tables of Hurst and Kavanagh (2017) and compared the results against the published values, cell by cell.

Selected-literature information

The full APA reference for the target article is Hurst and Kavanagh (2017) (Hurst & Kavanagh, 2017); Table 1 records its data source and the scope of the reproduction, following the course protocol.

Table 1

Table 1. Information on the selected article and the scope of the reproduction.

Item	Detail
Reference	Hurst and Kavanagh (2017); full entry in the reference list
DOI	10.1016/j.evolhumbehav.2016.06.001
Data source	Article supplementary files mmc1.csv and mmc2.xlsx (no OSF/GitHub)
Multiple studies / meta-analysis	No – single cross-sectional study
Previously replicated	None located
Reproduced results	All six results tables (descriptives, correlations, regressions)
Original raw data used	Yes – cleaned data file used unchanged
Sample size modified	No – full sample (N = 138; 92 women, 46 men)

Overview of the original study

Hurst and Kavanagh (2017) tested whether a faster life history strategy is accompanied by poorer mental-health and social outcomes. Participants completed two strategy measures (the Mini-K and the HKSS), the Adult Attachment Questionnaire (AAQ), the Buss–Perry Aggression Questionnaire, the DSM-5 Self-Rated Level 1 Cross-Cutting Symptom Measure, and a self-harm behaviour inventory, together with demographic and family-structure items. The authors reported descriptive statistics and reliabilities, the percentage of respondents exceeding each DSM-5 symptom threshold, self-harm frequencies by sex, correlations of all outcomes with age and sex, partial correlations between strategy and each outcome controlling for age (Bonferroni-corrected, $p < .002$), and regressions predicting each strategy score from attachment while controlling for age and sibling variables. They concluded that faster strategies were reliably linked to more psychopathology, more aggression, and poorer attachment, but not to self-harm

frequency or family structure—a pattern consistent with life history theory.

Method

Sample

I used the authors' supplementary data file (`mmc1.csv`, $N = 138$; 92 women, 46 men) together with the variable codebook (`mmc2.xlsx`). The data are cross-sectional and were taken unchanged; no cases were added or removed, and the sample size was not modified. Because the reproduction uses the identical sample, no power re-estimation was required.

Original analytic methods

The original study is cross-sectional and correlational. Reported analyses comprise means, standard deviations, and Cronbach's alpha; threshold percentages and frequency counts; zero-order Pearson correlations; first-order partial correlations controlling for age with a Bonferroni-corrected threshold; and ordinary-least-squares multiple regression with standardized coefficients and 95% confidence intervals. The software used by the original authors was not reported in the article. The present reproduction was conducted in R (R Core Team, 2024) using `psych` for descriptive statistics and reliability, `ppcor` for partial correlations, and `lm.beta` for standardized regression coefficients; the manuscript was prepared with `papaja` (Aust & Barth, 2023).

Replication approach

I reproduced all six tables of Hurst and Kavanagh (2017). The original authors' analysis code was not available, so the pipeline was written independently from the article's Method section and the codebook. Three implementation decisions were required and were resolved by reading the codebook rather than the article text. First, the DSM-5 items are stored on a 1–5 scale, whereas the published thresholds are defined on the

instrument’s native 0–4 scale; I subtracted one before thresholding. Second, the self-harm items are coded 1 = *yes*, 2 = *no*, so endorsements were counted as values equal to one rather than greater than zero. Third, the codebook showed that data columns and article rows are misaligned for self-harm items 5–7, so each behaviour was mapped to its correct column by name. For the “presence of stepparent” row I used the stepparent-count variable (`Stepparent_N`), which matches the published Table 5 values, rather than the binary indicator. Repeated operations—partial correlations, significance stars, and table assembly—were wrapped in functions so that each table required a single call. For every cell I recorded the published value, the reproduced value, a discrepancy index (δ), and a four-level rating: *fully replicated* (identical value and significance), *highly replicated* (last-digit rounding or a significance star only), *partially replicated*, and *not replicated*. The only methodological addition was a verification step in which the disputed Table 5 partial correlations were recomputed by a second, residual-based algorithm.

Results

Overall, computational reproducibility was high. Each of the following six tables places the published value (“Original result”) directly above the reproduced value (“This study”), with the discrepancy δ and the rating beneath. For the descriptive tables (Tables 2–4) δ is the relative deviation (percentage of the original); for the coefficient tables (Tables 5–8) δ is the absolute difference in the coefficient’s own units, because relative deviation is unstable near zero.

Descriptive statistics

Means, standard deviations, and reliability.

Table 2

Table 2. Comparison of original and replication: means, standard deviations, and reliability.

Measure / Statistic	M	SD	α
Total attachment (AAQ)			
Original result	15.54	7.63	0.92
This study	15.54	7.63	0.92
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Availability			
Original result	5.63	3.22	0.91
This study	5.63	3.22	0.91
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Goal-corrected partnership			
Original result	4.77	2.75	0.93
This study	4.77	2.75	0.93
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Angry distress			
Original result	5.14	2.84	0.83
This study	5.14	2.84	0.83
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Total aggression (Buss–Perry)			
Original result	59.90	19.62	0.93
This study	59.90	19.62	0.93
δ	0%	0%	0%

Table 2 continued

Measure / Statistic	M	SD	α
Rating	Fully repl.	Fully repl.	Fully repl.
Physical aggression			
Original result	16.81	6.04	0.83
This study	16.81	6.04	0.83
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Verbal aggression			
Original result	11.51	4.84	0.84
This study	11.51	4.84	0.84
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Anger			
Original result	14.30	5.44	0.83
This study	14.30	5.44	0.83
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Hostility			
Original result	17.27	8.05	0.90
This study	17.27	8.05	0.90
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.
Mini-K			
Original result	14.62	11.25	0.84
This study	14.62	11.25	0.84
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.

Table 2 continued

Measure / Statistic	M	SD	α
HKSS			
Original result	87.34	13.64	0.91
This study	87.34	13.64	0.91
δ	0%	0%	0%
Rating	Fully repl.	Fully repl.	Fully repl.

Note. α values are standardized Cronbach's α (psych::alpha std.alpha), which reproduced all 11 coefficients exactly; AAQ = Adult Attachment Questionnaire. All 33 cells fully replicated ($\delta = 0\%$).

114

115 Every mean, standard deviation, and reliability coefficient reproduced exactly. The
 116 reliabilities matched only once I used the *standardized* coefficient alpha (`std.alpha`); the
 117 raw alpha differed by .01–.02 on four scales. Because the article did not state which
 118 estimator it used, the agreement of all 11 standardized values indicates that the original
 119 authors reported standardized alpha. The descriptive layer of the article is therefore fully
 120 reproducible.

121 **DSM-5 symptom-threshold percentages.**

Table 3

Table 3. Comparison of original and replication: percentage meeting the DSM-5 threshold for further inquiry.

Item	Original	This study	δ	Rating
Depression subscale				

Table 3 continued

Item	Original	This study	δ	Rating
Little interest or pleasure	34.1	34.1	0%	Fully repl.
Feeling down/hopeless	29.7	29.7	0%	Fully repl.
Anger subscale				
More irritable/angry than usual	36.2	36.2	0%	Fully repl.
Mania subscale				
Sleeping less, still energetic	28.3	28.3	0%	Fully repl.
More projects/risky things	18.8	18.8	0%	Fully repl.
Anxiety subscale				
Nervous/anxious/on edge	35.5	35.5	0%	Fully repl.
Panic or frightened	21.0	21.0	0%	Fully repl.
Avoiding anxiety-provoking situations	34.1	34.1	0%	Fully repl.
Somatic symptoms subscale				
Unexplained aches and pains	29.0	29.0	0%	Fully repl.
Illness not taken seriously	15.2	15.2	0%	Fully repl.
Suicidal ideation subscale				
Thoughts of hurting yourself*	18.8	18.8	0%	Fully repl.
Psychosis subscale				
Hearing things others cannot*	7.2	7.2	0%	Fully repl.
Thought broadcasting/insertion*	6.5	6.5	0%	Fully repl.
Sleep disturbance subscale				
Sleep-quality problems	31.2	31.2	0%	Fully repl.
Memory subscale				
Memory/location problems	15.2	15.2	0%	Fully repl.
Repetitive thoughts/behaviour subscale				
Intrusive thoughts/urges/images	20.3	20.3	0%	Fully repl.
Compulsive acts	10.1	10.1	0%	Fully repl.

Table 3 continued

Item	Original	This study	δ	Rating
Dissociation subscale				
Feeling detached	13.8	13.8	0%	Fully repl.
Personality functioning subscale				
Not knowing who you are	30.4	30.4	0%	Fully repl.
Not feeling close to others	26.8	26.8	0%	Fully repl.
Substance use subscale				
≥ 4 drinks in a day*	34.1	34.1	0%	Fully repl.
Tobacco use*	12.3	12.3	0%	Fully repl.
Non-prescribed medicine use*	9.4	9.4	0%	Fully repl.

Note. Values are percentages of respondents whose rating indicated a need for further inquiry. *Threshold is a score of 1 rather than 2. Item labels are abbreviated. All 23 items fully replicated ($\delta = 0\%$).

122

123 All 23 threshold percentages reproduced to the first decimal. This table was,
124 however, the source of the first replication-side error: the DSM-5 items are stored on a 1–5
125 scale, but the published thresholds assume the instrument’s 0–4 scale, so applying the
126 thresholds directly inflated every percentage and pushed the special “score ≥ 1 ” items to
127 100%. Subtracting one before thresholding—a fact recoverable only from the
128 codebook—restored exact agreement.

129 **Self-harm frequencies by sex.**

Table 4

Table 4. Comparison of original and replication: frequencies (percentages) of self-harm behaviours by sex.

Measure / Statistic	Women n (%)	Men n (%)
1. Cutting		
Original result	13 (14.0)	6 (13.0)
This study	13 (14.1)	6 (13.0)
δ	+0.7%	0%
Rating	Highly repl.	Fully repl.
2. Burning with cigarette		
Original result	5 (5.4)	1 (2.2)
This study	5 (5.4)	1 (2.2)
δ	0%	0%
Rating	Fully repl.	Fully repl.
3. Burning with lighter/match		
Original result	3 (3.3)	3 (6.5)
This study	3 (3.3)	3 (6.5)
δ	0%	0%
Rating	Fully repl.	Fully repl.
4. Carving words into skin		
Original result	6 (6.5)	0 (0.0)
This study	6 (6.5)	0 (0.0)
δ	0%	0%
Rating	Fully repl.	Fully repl.
5. Carving pictures into skin		
Original result	1 (1.1)	0 (0.0)
This study	1 (1.1)	0 (0.0)
δ	0%	0%

Table 4 continued

Measure / Statistic	Women n (%)	Men n (%)
Rating	Fully repl.	Fully repl.
6. Severe scratching		
Original result	10 (10.9)	2 (4.3)
This study	10 (10.9)	2 (4.3)
δ	0%	0%
Rating	Fully repl.	Fully repl.
7. Biting		
Original result	6 (6.5)	2 (4.3)
This study	6 (6.5)	2 (4.3)
δ	0%	0%
Rating	Fully repl.	Fully repl.
8. Rubbing sandpaper on skin		
Original result	0 (0.0)	2 (4.3)
This study	0 (0.0)	2 (4.3)
δ	0%	0%
Rating	Fully repl.	Fully repl.
9. Dripping acid on skin		
Original result	0 (0.0)	0 (0.0)
This study	0 (0.0)	0 (0.0)
δ	0%	0%
Rating	Fully repl.	Fully repl.
10. Bleach/oven cleaner on skin		
Original result	0 (0.0)	0 (0.0)
This study	0 (0.0)	0 (0.0)
δ	0%	0%
Rating	Fully repl.	Fully repl.

Table 4 continued

Measure / Statistic	Women n (%)	Men n (%)
11. Sticking pins/needles into skin		
Original result	3 (3.3)	5 (11.4)
This study	3 (3.3)	5 (10.9)
δ	0%	-4.4%
Rating	Fully repl.	Partially repl.
12. Rubbing glass on skin		
Original result	1 (1.1)	0 (0.0)
This study	1 (1.1)	0 (0.0)
δ	0%	0%
Rating	Fully repl.	Fully repl.
13. Breaking bones		
Original result	1 (1.1)	0 (0.0)
This study	1 (1.1)	0 (0.0)
δ	0%	0%
Rating	Fully repl.	Fully repl.
14. Banging head		
Original result	1 (1.1)	5 (10.9)
This study	1 (1.1)	5 (10.9)
δ	0%	0%
Rating	Fully repl.	Fully repl.
15. Punching self		
Original result	4 (4.3)	2 (4.3)
This study	4 (4.3)	2 (4.3)
δ	0%	0%
Rating	Fully repl.	Fully repl.
16. Preventing wounds from healing		

Table 4 continued

Measure / Statistic	Women <i>n</i> (%)	Men <i>n</i> (%)
Original result	3 (3.3)	1 (2.2)
This study	3 (3.3)	1 (2.2)
δ	0%	0%
Rating	Fully repl.	Fully repl.
17. Alternate forms of self-harm		
Original result	4 (4.3)	8 (17.4)
This study	4 (4.3)	8 (17.4)
δ	0%	0%
Rating	Fully repl.	Fully repl.

Note. Women *n* = 92, men *n* = 46. δ is computed on the percentages.
All endorsement counts reproduced exactly; two percentages differ by
rounding with identical counts.

130

131 Every endorsement count reproduced exactly. Only two percentages deviate, and
132 both reflect rounding rather than a count error: cutting in women (14.0 vs. 14.1) and
133 sticking pins in men (11.4 vs. 10.9), each with an identical underlying count. Reaching this
134 point required two further codebook-based fixes: the self-harm items are coded 1 = *yes*, 2
135 = *no* (so endorsements are values equal to one, not greater than zero), and the data store
136 items 5–7 in a different order than the article lists them, so each behaviour had to be
137 matched to its column by name.

138 **Inferential statistics**

139 **Correlations with age and sex.**

Table 5

Table 5. Comparison of original and replication: zero-order correlations with age and sex.

Measure / Statistic	Age (r)	Sex (r)
Total attachment		
Original result	-.21*	.06
This study	-.21*	.06
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Availability		
Original result	-.15	.02
This study	-.15	.02
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Angry distress		
Original result	-.29**	.10
This study	-.29***	.10
δ	0.00	0.00
Rating	Highly repl.	Fully repl.
Goal-corrected partnership		
Original result	-.11	.03
This study	-.10	.03
δ	+0.01	0.00
Rating	Highly repl.	Fully repl.
Total aggression		
Original result	-.39***	.09
This study	-.39***	.09
δ	0.00	0.00

Table 5 continued

Measure / Statistic	Age (r)	Sex (r)
Rating	Fully repl.	Fully repl.
Physical aggression		
Original result	-.36***	.01
This study	-.36***	.00
δ	0.00	-0.01
Rating	Fully repl.	Highly repl.
Verbal aggression		
Original result	-.32***	-.06
This study	-.32***	-.06
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Anger		
Original result	-.26**	.19*
This study	-.26**	.19*
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Hostility		
Original result	-.32***	.13
This study	-.32***	.13
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Total psychopathology		
Original result	-.43***	.01
This study	-.43***	.01
δ	0.00	0.00
Rating	Fully repl.	Fully repl.

Table 5 continued

Measure / Statistic	Age (<i>r</i>)	Sex (<i>r</i>)
Total frequency of self-harm		
Original result	−.15	.01
This study	−.15	.02
δ	0.00	+0.01
Rating	Fully repl.	Highly repl.

Note. δ is the absolute difference (this study – original).

p* < .05. *p* < .01. ****p* < .001.

140

141 Every correlation coefficient reproduced to two decimals. The four highly replicated
142 cells differ only trivially: angry distress with age carries one extra significance star (.29**
143 vs. .29***), and three coefficients near zero shift by .01 in the last digit. None of these
144 changes the inferential message—the correlation is significant, or not, in exactly the same
145 way as in the article.

146 **Partial correlations with life history strategy.**

Table 6

Table 6. Comparison of original and replication: partial correlations (controlling for age) with Mini-K and HKSS.

Measure / Statistic	Mini-K (<i>pr</i>)	HKSS (<i>pr</i>)
Full biological siblings		
Original result	.01	.03
This study	.00	.03
δ	−0.01	0.00

Table 6 continued

Measure / Statistic	Mini-K (<i>pr</i>)	HKSS (<i>pr</i>)
Rating	Highly repl.	Fully repl.
Half-siblings		
Original result	−.10	−.20
This study	−.10	−.20
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Step-siblings		
Original result	−.05	−.08
This study	−.05	−.08
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Presence of stepparent		
Original result	−.19	−.06
This study	−.19	−.06
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Total psychopathology		
Original result	−.51	−.41
This study	−.51	−.41
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Total frequency of self-harm		
Original result	−.04	−.17
This study	−.04	−.17
δ	0.00	0.00
Rating	Fully repl.	Fully repl.

Table 6 continued

Measure / Statistic	Mini-K (<i>pr</i>)	HKSS (<i>pr</i>)
Total attachment		
Original result	−.42	−.41
This study	−.42	−.41
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Availability		
Original result	−.39	−.37
This study	−.38	−.37
δ	+0.01	0.00
Rating	Highly repl.	Fully repl.
Angry distress		
Original result	−.44	−.44
This study	−.44	−.44
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Goal-corrected partnership		
Original result	−.39	−.24
This study	−.27	−.24
δ	+0.12	0.00
Rating	Not repl.	Fully repl.
Total aggression		
Original result	−.46	−.43
This study	−.44	−.43
δ	+0.02	0.00
Rating	Highly repl.	Fully repl.
Physical aggression		

Table 6 continued

Measure / Statistic	Mini-K (<i>pr</i>)	HKSS (<i>pr</i>)
Original result	−.44	−.34
This study	−.37	−.34
δ	+0.07	0.00
Rating	Partially repl.	Fully repl.
Verbal aggression		
Original result	−.03	−.10
This study	−.04	−.10
δ	-0.01	0.00
Rating	Highly repl.	Fully repl.
Anger		
Original result	−.31	−.25
This study	−.31	−.25
δ	0.00	0.00
Rating	Fully repl.	Fully repl.
Hostility		
Original result	−.52	−.52
This study	−.52	−.52
δ	0.00	0.00
Rating	Fully repl.	Fully repl.

Note. Bold = significant after Bonferroni correction ($p < .002$). δ is the absolute difference. Two Mini-K cells could not be reproduced (goal-corrected partnership, physical aggression); their HKSS counterparts reproduced exactly (see text).

147

148 The HKSS column reproduced exactly in all 15 rows, and 13 of the 15 Mini-K
149 coefficients reproduced within rounding. The exceptions are the two Mini-K cells flagged in
150 bold above: goal-corrected partnership (article $-.39$, data $-.27$) and physical aggression
151 (article $-.44$, data $-.37$). These are best read as transcription errors in the original article
152 rather than replication failures, for three reasons: the corresponding HKSS coefficients
153 reproduced perfectly, a second residual-based partial-correlation algorithm returned the
154 same disputed values, and both cells remain significant after Bonferroni correction under
155 either number—so the article’s pattern of significant strategy–outcome associations is
156 preserved regardless.

157 **Regressions predicting life history strategy.**

Table 7
Table 7. Comparison of original and replication: regression predicting Mini-K from attachment.

Measure / Statistic	b	β	t	CI LL	CI UL
Age (covariate)					
Original result	0.21	.27***	3.29	0.08	0.34
This study	0.21	.27**	3.29	0.08	0.34
δ	0.00	0.00	0.00	0.00	0.00
Rating	Fully repl.	Highly repl.	Fully repl.	Fully repl.	Fully repl.
Half-siblings (covariate)					
Original result	−0.01	.01	−0.01	−2.16	2.16
This study	0.00	.00	0.00	−2.16	2.16
δ	+0.01	−0.01	+0.01	0.00	0.00
Rating	Highly repl.	Highly repl.	Highly repl.	Fully repl.	Fully repl.
Step-siblings (covariate)					
Original result	0.21	.02	0.19	−1.93	2.36

Table 7 continued

Measure / Statistic	b	β	t	CI LL	CI UL
This study	0.21	.01	0.19	−1.93	2.36
δ	0.00	−0.01	0.00	0.00	0.00
Rating	Fully repl.	Highly repl.	Fully repl.	Fully repl.	Fully repl.
Biological siblings (covariate)					
Original result	−0.04	−.01	−0.05	−1.69	1.61
This study	−0.04	.00	−0.05	−1.69	1.61
δ	0.00	+0.01	0.00	0.00	0.00
Rating	Fully repl.	Highly repl.	Fully repl.	Fully repl.	Fully repl.
Total attachment					
Original result	−0.60	−.41***	−5.20	−0.83	−0.37
This study	−0.60	−.41***	−5.20	−0.83	−0.37
δ	0.00	0.00	0.00	0.00	0.00
Rating	Fully repl.	Fully repl.	Fully repl.	Fully repl.	Fully repl.

Note. b = unstandardized; β = standardized; CI = 95% confidence interval (LL, UL).

Model: $R^2 = .28$, $F(5, 132) = 10.49$, $p < .001$, reproduced exactly. * $p < .05$. ** $p < .01$.

*** $p < .001$.

Table 8

Table 8. Comparison of original and replication: regression predicting HKSS from attachment.

Measure / Statistic	b	β	t	CI LL	CI UL
Age (covariate)					
Original result	0.20	.21*	2.51	0.04	0.35
This study	0.20	.21*	2.51	0.04	0.35
δ	0.00	0.00	0.00	0.00	0.00

Table 8 continued

Measure / Statistic	b	β	t	CI LL	CI UL
Rating	Fully repl.	Fully repl.	Fully repl.	Fully repl.	Fully repl.
Half-siblings (covariate)					
Original result	-1.63	-.10	-1.21	-4.30	1.04
This study	-1.63	-.10	-1.21	-4.30	1.04
δ	0.00	0.00	0.00	0.00	0.00
Rating	Fully repl.	Fully repl.	Fully repl.	Fully repl.	Fully repl.
Step-siblings (covariate)					
Original result	0.07	.01	0.06	-2.58	2.72
This study	0.07	.00	0.06	-2.58	2.72
δ	0.00	-0.01	0.00	0.00	0.00
Rating	Fully repl.	Highly repl.	Fully repl.	Fully repl.	Fully repl.
Biological siblings (covariate)					
Original result	0.01	.01	0.01	-2.03	2.05
This study	0.01	.00	0.01	-2.03	2.05
δ	0.00	-0.01	0.00	0.00	0.00
Rating	Fully repl.	Highly repl.	Fully repl.	Fully repl.	Fully repl.
Total attachment					
Original result	-0.67	-.37***	-4.68	-0.95	-0.38
This study	-0.67	-.37***	-4.68	-0.95	-0.38
δ	0.00	0.00	0.00	0.00	0.00
Rating	Fully repl.	Fully repl.	Fully repl.	Fully repl.	Fully repl.

Note. Layout as in Table 7. Model: $R^2 = .26$; $F(5, 132) = 9.09$ original vs. 9.08 replication (last-digit difference, highly replicated). * $p < .05$. *** $p < .001$.

Table 9

Table 9. Distribution of reproduction ratings across all 196 paired comparisons.

Reproduction rating	<i>N</i>	%
Fully replicated ($\delta = 0$)	175	89.3
Highly replicated (last-digit rounding or significance star only)	18	9.2
Partially replicated	2	1.0
Not replicated	1	0.5
Total	196	100.0

Note. The 18 highly replicated cells reflect last-digit rounding or one shifted significance star. The 2 partially and 1 not replicated cells are the two suspected Mini-K transcription errors in the partial-correlation comparison (Table 6) and one rounded self-harm percentage with an identical count.

162 effect ($\beta = -.41$ for Mini-K and $-.37$ for HKSS, both $p < .001$) reproduced exactly. The
163 handful of highly replicated cells are standardized coefficients near zero that differ by .01,
164 plus the HKSS model F (9.09 vs. 9.08). The model fit statistics— $R^2 = .28$ and
165 .26—reproduced exactly. No regression inference changed.

166 **Assessment of computational reproducibility**

167 I treated every published number as one paired comparison: each mean, standard
168 deviation, reliability, percentage, frequency, correlation, partial correlation, regression
169 coefficient, confidence-interval limit, and model-fit statistic. This yielded 196 comparisons,
170 whose rating distribution appears in Table 9.

171 Reproducibility of the published *numbers* is distinct from reproducibility of the
172 *inferences* they license. Counting every significance decision—22 zero-order correlations, 30

Table 10
*Table 10. Consistency of
statistical inferences
between original and
replication.*

Inference	<i>N</i>	%
Consistent	64	100.0
Inconsistent	0	0.0
Total	64	100.0

Note. An inference was coded inconsistent only if the reproduced p value fell on the opposite side of its significance threshold from the original.

173 age-adjusted partial correlations, and 12 regression statistics (10 coefficient tests plus two
174 model *F* tests)—gives 64 inferences. As Table 10 shows, all 64 fell on the same side of their
175 significance threshold in the original and the replication; even the two unreproduced
176 Mini-K point estimates stayed significant under both values.

177 Because the reproduction used the original methods throughout, the rubric’s tables
178 for an alternative (“innovative”) analysis method are not applicable; the only
179 methodological addition was the residual-based cross-check of Table 6, which agreed with
180 the primary estimates.

Discussion

Analysis of the reproducibility outcome

Using the authors’ open data, the central results of Hurst and Kavanagh (2017) were highly reproducible. Because the authors archived a *cleaned* data file, the reproduction succeeded almost completely: means, standard deviations, reliabilities, correlations, partial correlations, and regression coefficients matched the published values to the last reported digit in the large majority of cells, and no inference supporting the article’s conclusions changed. The few small deviations that remained—a scattering of last-digit differences and a single shifted significance star—are attributable to rounding and to unspecified estimator options rather than to any computational instability, and none affects an inference. In short, the article is computationally robust and trustworthy: the numbers it reports can be regenerated from its data, and its conclusions follow from those numbers.

The residual discrepancies sort into three kinds, summarized in Table 11, which it is important not to conflate. *First*, two Mini-K partial correlations in Table 6 (goal-corrected partnership and physical aggression) could not be reproduced; these are most plausibly transcription errors in the original article, because the matching HKSS coefficients reproduced exactly, two independent algorithms agreed, and both cells remain significant either way. *Second*, three discrepancies that initially made the DSM-5 and self-harm tables look completely irreproducible were located entirely on the replication side—a scoring-origin mismatch (1–5 vs. 0–4), a yes/no coding error, and an item-to-column misalignment—and disappeared once the codebook was consulted. *Third*, the benign remainder (last-digit rounding, one rounded self-harm percentage with an identical count, and the standardized-versus-raw alpha choice) reflects rounding and reporting conventions. The principal limitation to full transparency was structural rather than statistical: the raw and cleaned data were shared, but the analysis code was not, so a few implementation choices—most visibly the alpha estimator—had to be inferred. Sharing runnable code

Table 11

Table 11. Analysis of the sources of the few computational discrepancies.

Possible source	Applies?	Note
Transcription error in original	Yes	Two Mini-K cells in Table 6
Under-specified procedure in article	Partly	Coding and alpha from codebook only
No analysis code shared	Yes	Main openness gap; data shared, code not
Software/package versions not reported	Yes	Minor; one last-digit F difference
Same data set	No	Identical cleaned file used
Different software (R)	Minor	Rounding / alpha estimator only
Replicator’s reading of data coding	Resolved	Three coding errors fixed via codebook
Publication year	–	2017, open-science era; data archived
Independent support for findings	Yes	Pattern broadly consistent with prior work

Note. Only applicable rows are shown; non-applicable checklist items (e.g., unseeded simulation – all analyses are deterministic; OSF storage or download problems) are omitted.

207 would have removed that ambiguity.

208 **Further reflections**

209 Beyond the reproducibility verdict, three lessons stand out from the process.

210 *Functionalizing the analysis sharply lowered the error rate.* Wrapping the repeated
211 operations—computing a partial correlation, attaching significance stars, and assembling a
212 table—into reusable functions meant that each of the six tables was produced by a single
213 call pattern (the `blk()` helper used throughout the Results section is a direct example).
214 This made discrepancies easy to localize and guaranteed that the same
215 conventions—rounding, star thresholds, ordering—were applied everywhere. That

uniformity is difficult to maintain in point-and-click software, where each table is built by hand, and it is a core advantage of a scripted environment such as R.

The data codebook mattered more than the code. The single most important lesson was that all three errors encountered during the reproduction lay not in the statistical method but in how the raw data were encoded: whether scoring began at 0 or at 1, how the yes/no self-harm items were coded, and whether the data's column order matched the article's row order. However well written the R code was, a misunderstanding of the data's semantics propagated into a systematic departure from the published values. Reading the codebook closely *before* reading the data should be a default habit, not an afterthought.

Working with an AI assistant requires verification, not trust. Much of the code was written in dialogue with Claude. The decisive practice was not to take its output on faith but to think alongside it—cross-checking each result against the original article and the data through repeated exchanges, especially for the values to be reproduced and for every data-processing step. In particular, the moment a suggested script ran without error was precisely the moment to slow down and inspect it carefully: a clean run is not evidence of a correct answer. Synchronizing human judgment with the assistant's speed, rather than substituting one for the other, is what kept the reproduction honest.

Conclusion

The core results of Hurst and Kavanagh (2017) are computationally reproducible from the authors' open data: 89.3% of values matched exactly, a further 9.2% differed only by rounding, and every inference was preserved. The two unreproducible cells are best explained as typographical errors in the original article, and the discrepancies that first appeared were, once diagnosed, properties of the replication code rather than of the study. Robust computational reproducibility rests on two pillars—the original authors' openness and the replicator's disciplined attention to how the data are encoded—and, increasingly,

241 on treating AI-assisted code as something to be checked rather than believed.

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